

EXPERIMENT 12 – Website Traffic Analysis

Question

Analyze website traffic using line chart, bar chart, and table.

Dataset

Date	Page Views	Click-through Rate
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2023-01-01	1500	2.3%
------------	------	------

2023-01-02	1600	2.7%
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2023-01-03	1400	2.0%
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The supplied experiment specifies these three days and asks for traffic visualization.

R Code

```
# Experiment 12 - Website Traffic
```

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
```

```
page_views <- c(1500, 1600, 1400)
```

```
ctr <- c(2.3, 2.7, 2.0)
```

```
data <- data.frame(date, page_views, ctr)
```

```
# Question 1 - Line Chart
```

```
plot(date, page_views,
```

```
  type = "o",
```

```
  main = "Daily Page Views",
```

```
  xlab = "Date",
```

```
  ylab = "Page Views")
```

```
# Question 2 - Bar Chart for Top N CTR
```

```
barplot(ctr,
```

```
  names.arg = format(date, "%d-%b"),
```

```
  main = "Click-through Rate",
```

```
xlab = "Date",  
ylab = "CTR (%)",  
col = "lightgreen")
```

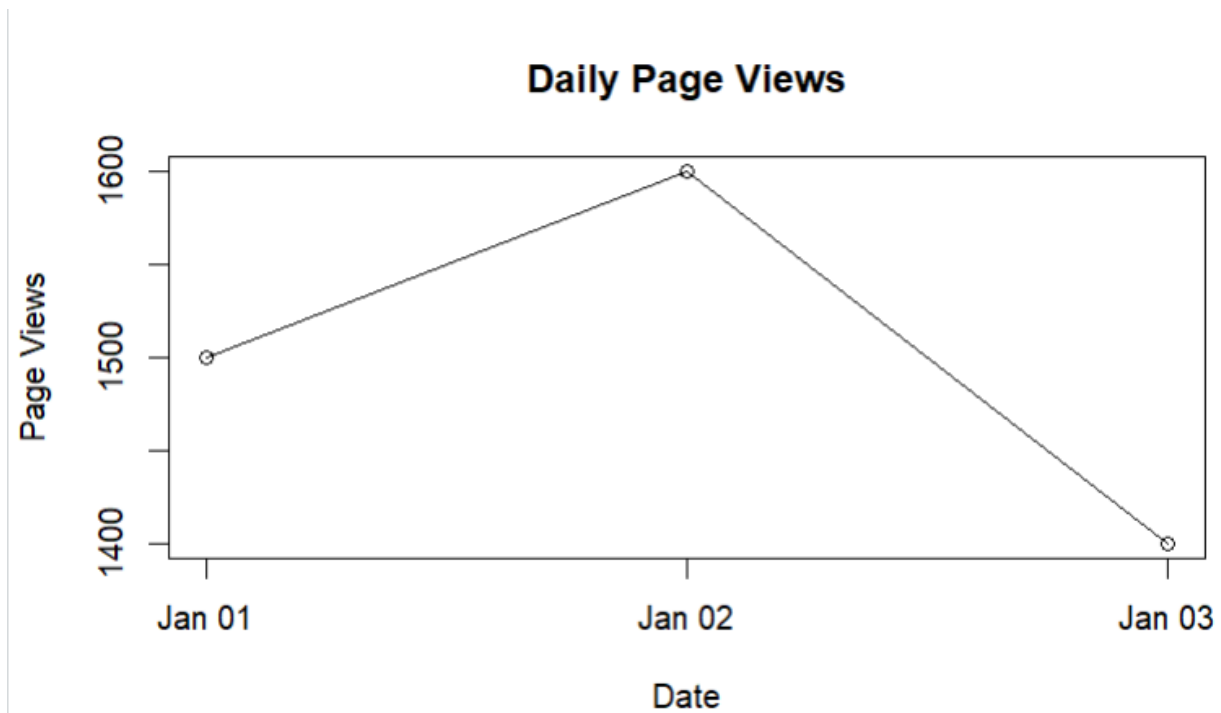
Question 3 - Table

```
print(data)
```

Question 1

Create a line chart to visualize the trend in daily page views over time.

Output 1

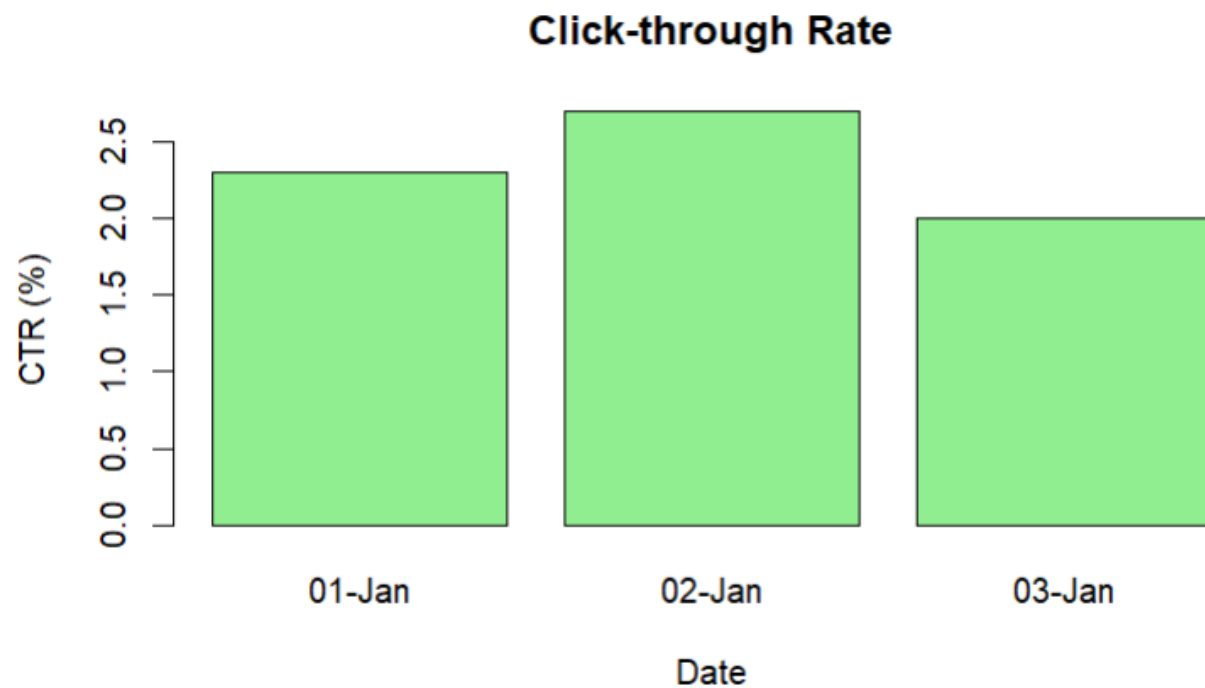


The line chart shows page views increasing from **1500 to 1600**, followed by a decrease to **1400**.

Question 2

Generate a bar chart showing the days with the highest click-through rates.

Output 2



The highest click-through rate is on **2023-01-02** with **2.7%**.

Question 3

Build a table showing website traffic data.

Output 3

```
> # Question 3 - Table
> print(data)
      date page_views ctr
1 2023-01-01      1500 2.3
2 2023-01-02      1600 2.7
3 2023-01-03      1400 2.0
```